26

- **Title:** *An IoT-based QR-code-enabled smart medicine box for reducing prescription errors and improving medication adherence*
- **Most important information:**
  - The prescription itself is a QR code: the doctor encodes it, the box reads it. That takes handwriting out of the loop, which none of the others try to do
  - ESP32 + RTC, Wi-Fi to a cloud database
  - *Dispensing:* none. Multi-drawer box, one compartment per medicine, patient opens whichever drawer is lit
  - *Intake verification:* reed switch on each drawer. Drawer opens = dose taken, that is the whole check
  - Best numbers of anything I read: against a manual pill box over 40 events, adherence 75% → 97.5%, missed doses 7 → 1, interpretation errors 20% → 0%. 98% drawer detection over 50 triggers
- **Limitations:**
  - Drawer opening is treated as intake
  - LED and LCD only, no buzzer, and no caregiver alerts (they list those as future work)
  - Needs internet to pull new prescriptions
  - They never say how many drawers there are, and give no participant count for the 40-event test
  - Manual refilling, and the medicine comes out of its packaging

27

- **Title:** *IoT-enabled smart pill dispenser for enhanced medication compliance and safety*
- **Most important information:**
  - Arduino UNO + RTC, GSM (SIM800L) for SMS, no Wi-Fi and no app
  - *Dispensing:* an SG90 servo turns a tray and drops one dose; a solenoid lock keeps everything shut outside dosing times, which is their overdose prevention
  - *Intake verification:* IR sensors at the slot, the pill has to be picked up within about 10 s or it counts as missed
  - Escalation is decent: buzzer and LCD first, then alarm plus SMS to patient and caregiver, delivered in 5–10 s
- **Limitations:**
  - Breadboard only, and no adherence numbers
  - Their own list: IR gets confused in bright light, schedules have to be reprogrammed by hand in the code, compartments too small for complex regimens, depends on GSM coverage, no power backup
  - The abstract mentions an app and “favorable user response”, but there is no app and no user study anywhere in the paper
  - Manual refilling, and the pills have to be loose since the servo drops them one at a time

28

- **Title:** *MedicBox: Smart pill reminder box*
- **Most important information:**
  - ESP32 + DS3231, Wi-Fi to Firebase, and it caches the schedule so it still works offline
  - *Dispensing:* none, 7 compartments opened by hand
  - *Intake verification:* two signals, both needed. A microswitch logs “Box Accessed”, then the patient taps “Confirm Taken” in the app
  - Caregiver side is the most developed I saw: three roles (self-use, caregiver-only, joint), remote schedule setup, a log that separates Accessed / Confirmed / Missed, and one caregiver can follow several patients
  - Flutter app, Android and iOS
- **Limitations:**
  - Functional test only, zero numbers. The abstract claims sub-second sync but never shows it
  - They say themselves they have no load cells, so partial doses are invisible. Wi-Fi only as well
  - A lid opening still does not tell you how many pills came out, and the second check needs the patient to use a phone app
  - Manual refilling

29

- **Title:** *Pill bottle mounted wireless sensor using IoT*
- **Most important information:**
  - ESP32, one sensor per compartment, LCD, LEDs and buzzer, 2 open compartments
  - *Dispensing:* none
  - *Intake verification:* compartment access, nothing more. To their credit they say so outright, and list how the sensors can be fooled by placement, handling or accidental triggering
- **Limitations:**
  - No experiment at all. They state this themselves and just list metrics someone should measure later
  - Never say what kind of sensor it is, and say nothing about power
  - Wi-Fi and caregiver notifications are described as possible, not built
  - Title says pill bottle mounted, but the prototype is not mounted on a bottle

30

- **Title:** *PillMediCount: An affordable smart IoT device for medication tracking*
- **Most important information:**
  - Closest thing to real dose counting I found: 3 TAL221 load cells with HX711 amplifiers, one under each tray
  - *Intake verification:* tray weight divided by the reference pill weight gives the count, so pills coming out show up as the number dropping. Better than lid sensing, though they never compare that drop against the schedule
  - Arduino Uno, Bluetooth HC-06 to an Android app with a local SQLite database, no cloud at all
  - Nothing on the device itself: no LED, no buzzer, no screen. Reminders and refill alerts all live in the app

    - Only one with a proper cost analysis ($44.99 against $89.99–$169 commercial), and one of only two that actually states a battery (9 V Li-ion, about 2 weeks)
- **Limitations:**
    - Counting gets worse as the tray fills: exact up to about 24 pills, then ±2 between 25 and 30
    - One pill weight tested (0.8 g), bench only, no users
    - No alert on the device, no missed-dose logic, no caregiver, 3 medicines maximum
    - Manual refilling

31

- **Title:** *Smart medicine reminder box for healthcare monitoring using IoT*
- **Most important information:**
    - ESP32, Wi-Fi, takes the time from NTP instead of an RTC
    - *Dispensing:* 3 servos, one per time slot, and the software only lets one dose out per slot
    - *Intake verification:* none. The patient presses a button or taps the app. They mention a "dual-verify logic" and never explain what it is
    - Remote side is the most complete of the lot: patient / caregiver / doctor / maintainer roles, schedule editing, adherence reports, even remote firmware updates
    - 99/100 dispensing cycles, 98% alerts, and it kept logging when the cloud connection dropped
- **Limitations:**
    - No intake sensing at all, so the whole adherence record is self-report
    - Those cycles were simulated, plus a Proteus simulation. The photo in the paper is bare components on a bench, no box
    - They mention a user evaluation but give no method, no participants and no numbers
    - 60 s to acknowledge before it marks a miss is short for elderly users

32

- **Title:** *Smart pill dispenser with facial recognition and health monitoring*
- **Most important information:**
    - Only one that checks who you are before dispensing: the servos release only after face recognition (LBPH, threshold 60)
    - ESP32 + camera, plus MAX30102, LM35 and a BP sensor, so it doubles as a vitals monitor
    - *Intake verification:* none
- **Limitations:**
    - Design and simulation only, nothing was built. Face recognition and missed-dose accuracy cannot come out of a simulation
    - The numbers do not even agree with each other between sections: dispensing 98.7 / 98.5 / 98.8%, face recognition 95 / 96 / 94.7%
    - The comparison table has no source
    - Face recognition at 20–25 fps on an ESP32 seems unlikely for that chip
    - Nothing about the physical box, the compartments or power
    - Probably the weakest of my twelve, so worth dropping if we are short on space

33

- **Title:** *Medication adherence and liquid level tracking system for healthcare provider feedback*
- **Most important information:**
  - The one that leaves the medicine where it is: a 3D-printed sleeve clips around the original prescription bottle without changing it, and fits different bottle sizes
  - nRF51422 SoC, BNO080 IMU, two reed switches for cap on/off, and an FDC1004 capacitive sensor for fluid level and squeeze force
  - *Intake verification:* by far the best I read. A random forest running on the device uses orientation and squeeze pattern to decide whether a drop actually came out
  - Properly evaluated too: 10 adults aged 65+, 100 real uses against 100 cap-off-without-use events, plus deliberate fake uses. TPR 0.97, FPR 0.11, F1 0.93. Fluid level $R^2 = 0.998$
  - Also one of the few that bothers with power (85 mAh, about 17.5 h) and cost (~$50 a unit against ~$200/month commercial)
- **Limitations:**
  - Eye drops, so the sensing does not carry over to tablets
  - Still gets false positives when someone fakes a use
  - 17 h of battery, and the sleeve makes the bottle stiffer to squeeze
  - Lab only, and the data goes to the clinic rather than to a caregiver

34

- **Title:** *Unobtrusive patient identification using smart pill-bottle systems*
- **Most important information:**
  - Different question from everyone else: who is taking the pills, worked out from hand movement. Aimed at opioid misuse in a household rather than at adherence
  - Two accelerometers (bottle and cap) at 10 Hz and a magnetic cap switch, RF back to a PC
  - *Intake verification:* not their problem, they do not attempt it
  - Biggest study I came across: 36 healthy adults, candy instead of pills. 90% for one user out of 3, down to 85% and 72% recall once there are 16
- **Limitations:**
  - Young, healthy, mostly male participants, candy, lab
  - Falls apart as the household gets bigger
  - Says nothing about whether the pill was taken, and needs a PC as base station
  - Custom printed bottle, not a real pharmacy one

35

- **Title:** *An IoT-based pill management system for elderly*
- **Most important information:**
  - Reminder flow is the closest to ours: phone notification 10 minutes before, then LED, buzzer and LCD when it is actually time
  - Arduino Uno + ESP8266, Android app with four roles and an “ask for help” button
  - *Dispensing:* a stepper turns the container to the right compartment, and the servo only unlocks it once the patient has confirmed they are there

- *Intake verification:* none. Touch button for presence, touch again for intake
- Only paper with a pharmacist in the workflow: the pharmacist fills the compartments and sets the schedule through the app, while the doctor edits prescriptions remotely

- **Limitations:**
  - 7 people, 3 days each, and only satisfaction feedback. No numbers
  - No intake sensor, and few compartments
  - The database was not even connected yet when they wrote it
  - Manual refilling

36

- **Title:** *Intelligent and assisted medicine dispensing machine for elderly visual impaired people with deep neural network fingerprint authentication system*
- **Most important information:**
  - The only one I found where the medicine stays in its blister strip. The user pulls the strip out, pops the pills, puts the strip back
  - *Intake verification:* a load cell under each chamber weighs the strip before and after, so they compare what was removed against the prescribed dose (±20%) and flag over- or under-dosing. Most direct dose check in anything I read
  - 3 Raspberry Pis plus a PC, fingerprint scanner for access, two ultrasonic sensors per chamber to guide the hand to the right one
  - Built around accessibility: text-to-speech through the app, Braille numbers on the chambers, and it repeats the announcement if the user has not come to the box after about 7 min
  - Refill alert to pharmacist and doctor below 4 tablets. Ran for 3 months in a patient's home
- **Limitations:**
  - One patient. And the home results are just example tables, no adherence figures
  - The 98.3% fingerprint accuracy comes from SOCOFing, where the task is telling a thumb from an index finger, not telling people apart. So it does not show what they say it shows
  - Huge: a PC and three Raspberry Pis
  - They admit there is no power-outage handling, no fall detection and no secure messaging

37

- **Title:** *Design of a smart medical box for automatic pill dispensing and health monitoring*
- **Most important information:**
  - Raspberry Pi, R307 fingerprint sensor, MAX30102, DS18B20, HC-SR04
  - *Dispensing:* 3 steppers with single-pill disks, releases only after a fingerprint match. The disks are swappable for different pill sizes, which is a neat idea
  - *Intake verification:* none
  - GSM SMS to patient and doctor, functional test with 13 people aged 20–67
- **Limitations:**
  - No real metrics, just that everything worked and the data saved
  - They admit the disk can drop more than one pill, which is a problem when the whole design is single-pill dispensing

- Tablets only, 3 medicines, no app, no intake check
- Manual refilling